

Dinesh Barri

Dublin, Ireland | +353 89 203 3146 | dineshbarri1997@gmail.com
linkedin.com/in/dinesh-barri | github.com/dineshbarri | dineshbarri.dev

Professional Summary

AI / Data Engineer with **5+ years** building production GenAI systems, ML pipelines, and large-scale data analytics across enterprise environments. Proven expertise in **Python, SQL, LLMs, RAG, Vertex AI, Gemini, BigQuery, and PyTorch** — designing agentic workflows, ETL pipelines, and deployed ML applications end-to-end. Delivered AI automation and analytics platforms reducing manual effort by **70%**, processing **millions of records** daily, and achieving **98%+** model accuracy in production. Bridges the full stack from data modelling and ML engineering to GenAI product delivery and stakeholder-ready insights at scale.

Work Experience

Plemdo AI – Freelance Analytics & Automation Consulting

Dublin, Ireland

Freelance AI / Data Engineer & GenAI Automation Consultant

Oct 2024 – Present

- Architected production AI agent workflows using **Gemini APIs** and n8n, with structured outputs, retry logic, and validation for enterprise-grade reliability.
- Engineered end-to-end ETL pipelines covering data ingestion, cleansing, transformation, KPI generation, and automated multi-channel BI report delivery.
- Deployed GenAI reporting solutions integrating **RAG**, prompt engineering, and context management, reducing client manual reporting effort by **70%**.
- Built data validation and exception alerting layers across pipelines, ensuring production-quality outputs for SaaS and professional services clients.
- Delivered reusable deployment documentation covering OAuth, API configuration, and workflow onboarding, cutting client setup time by **40%**.

Cognizant – Google Maps Operations

Hyderabad, India

Data Analyst / Reporting Analyst

Jun 2019 – Jun 2023

- Analysed large-scale **Google Maps** operational datasets processing **millions of records** daily using **SQL, BigQuery, Python**, and Looker.
- Engineered **10+** automated KPI dashboards and ETL pipelines in BigQuery and Looker Studio, reducing manual reporting effort by **60%** across teams.
- Automated data validation and reconciliation checks across **BigQuery** pipelines, improving business-critical reporting accuracy by **30%**.
- Designed scalable data models and KPI frameworks in **SQL**, translating complex operations requirements into executive insight packs for leadership.
- Partnered with engineering teams to diagnose and resolve data pipeline defects, improving consistency and reliability across global reporting cycles.
- Delivered cross-functional analytics supporting product, operations, and trust teams across a **Google Maps** global data programme at enterprise scale.

ATHARVO

Bangalore, India

Machine Learning Engineer Intern

Feb 2019 – May 2019

- Developed supervised ML pipelines in **Python** using scikit-learn, pandas, and TensorFlow, improving experiment reproducibility by **25%**.
- Implemented data preprocessing, feature engineering, model training, and error analysis across classification and regression prediction tasks.
- Applied statistical evaluation and cross-validation techniques to benchmark model performance across structured operational datasets.

Technical Skills

- AI & GenAI Engineering:** LLMs, Generative AI, Prompt Engineering, RAG, Agentic Workflows, Multi-Agent Orchestration, Context Management, AI Evaluation, Gemini, Vertex AI, LangChain, n8n
- Data Engineering:** Python, SQL, ETL/ELT Pipelines, BigQuery, Snowflake, Airflow, dbt, Spark, Kafka, REST APIs, JSON, Data Validation, Data Modelling, Data Governance
- ML Engineering:** PyTorch, TensorFlow, scikit-learn, pandas, NumPy, CNN, Model Training & Evaluation, Feature Engineering, ML Lifecycle, Experiment Tracking
- Analytics & Applications:** Power BI, Looker Studio, Excel, Flask, FastAPI, Streamlit, GitHub, Docker, Google Cloud Platform, KPI Reporting, GDPR Awareness

Projects

- Plemdo AI Enterprise Analytics – Conversational BI Platform** | [GitHub](#) *n8n, Gemini, Python, APIs*
- Built an agentic AI analytics platform converting WhatsApp business queries into executive BI reports via **Google Gemini** and REST APIs.
 - Designed full production workflow: intent classification, AI-driven data analysis, dynamic **QuickChart** visualisation, PDF generation, and automated delivery.
 - Integrated prompt engineering, **RAG** context handling, and structured output validation, ensuring governance-grade reliability for enterprise decision-making.
- AI Video Factory – GenAI Automation Pipeline** | [GitHub](#) *Gemini, Vertex AI, n8n, Python*
- Architected a 6-stage production GenAI pipeline automating **Veo3** video creation, metadata tagging, Google Drive storage, and YouTube publishing.
 - Implemented **Vertex AI** API orchestration with structured prompts, retry logic, failure alerting, and audit logging; earned **16 GitHub stars**.
 - Documented end-to-end monitoring, recovery paths, and deployment guides enabling scalable extension across new GenAI automation workflows.
- Neural Digit Recognizer – Production ML Web App** | [GitHub](#) *PyTorch, CNN, Flask, Python*
- Deployed a **PyTorch** CNN application for real-time digit recognition via a Flask REST API, achieving **98%+** accuracy on the MNIST validation set.
 - Delivered full ML lifecycle: data preprocessing, CNN architecture design, model training, REST API serving, and browser-based front-end deployment.

Education

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| Dublin Business School
<i>MSc Data Science and Analytics — Full Time, 2:1 Honours</i> | Dublin, Ireland
<i>Sep 2023 – Sep 2024</i> |
| Gayatri Vidya Parishad College of Engineering
<i>B.Tech Computer Science and Engineering</i> | India
<i>Apr 2015 – Apr 2019</i> |

Certifications

- | | |
|---|---------------------------|
| Generative AI Leader Specialization | Google Cloud Training |
| Gemini for Data Scientists and Analysts | Google |
| Prepare Data for ML APIs on Google Cloud | Google Cloud Skills Boost |